

Lab 2: Programming I/O in Linux

Working with file

1. Given file *Hello.txt* with content: "Hello! What is your name?". Write a program to:
 - a) Read the content of file Hello.txt and print on the screen.
 - b) Read the content of file Hello.txt starting from the 7th byte from the beginning of the file and print on the screen. (*Hint*: using lseek system call)
2. Write a program that creates a file with two options:
Select 1: The contents of the file are entered from the keyboard.
Select 2: Read the contents of another file and write it to the file you just created.

Optional: Create two versions of above program: one version using basic system call *open()*, *read()*, *write()*, *close()* and one version using standard C library. Compare time needed to executing two versions of program? (You can use *time* shell command) Which one is faster? Why?

3. Write a program to enter the name of a file and print out the size of the file, the access mode information of the file. (*Hint*: you can use stat or fstat)
4. Write a program named linuxecho.c emulator:
echo [string] > example.txt.
(Printed string is placed in argv[1], filename is placed in argv[2])
5. Write a program called linuxgrep.c that emulates "grep" so that the filtered word is placed in argv[1], and the path to the file is placed in argv[2].
6. (*Optional*) Write a program named grepdir.c that list all the files in the folder containing the filtered word. The program receives 2 input parameters: filtered words in argv[1] and links to folders in argv[2].
Hint: Browse the input directory, for each file in the directory, reuse the linuxgrep program code in exercise 5 to filter the content of the file with the filtered word. If the filtered word appears in the file, print the file name.

Working with directory

7. Write a program linuxls.c that emulates **ls** shell command. The program receives 1 input parameters: directory path in argv[1].
8. Write a program linuxcd.c that emulates **cd** shell command. The program receives 1 input parameters: directory path in argv[1]. (*Hint*: using chdir system call)